

Supersonic PINN vs validated modal CSV

Run: assets/pinn_supersonic/singlecase_pq_firstorder_M150_a01625_ASYMTOTICENV_lowtail_ft

alpha: 0.1625

Mach: 1.5

p_rel: 0.00948884

q_rel: 0.0109218

p_y_num_rel: 0.0267696

q_vs_dpred_rel: 0.0228544

align_scale_abs: 0.443741

pred_max_abs_p_raw: 1.00229

Zone errors:

center_|y|<=50: p_rel=0.003964, q_rel=0.006824, max|p_pred|=0.4448

center_|y|<=60: p_rel=0.009948, q_rel=0.008026, max|p_pred|=0.4448

diagnostic_|y|<=80: p_rel=0.02702, q_rel=0.02409, max|p_pred|=0.4448

left_-120_-80: p_rel=0.3659, q_rel=0.5196, max|p_pred|=0.1941

right_50_120: p_rel=1.071, q_rel=3.733, max|p_pred|=0.02539

right_y>80: p_rel=0.8336, q_rel=2.555, max|p_pred|=0.003864

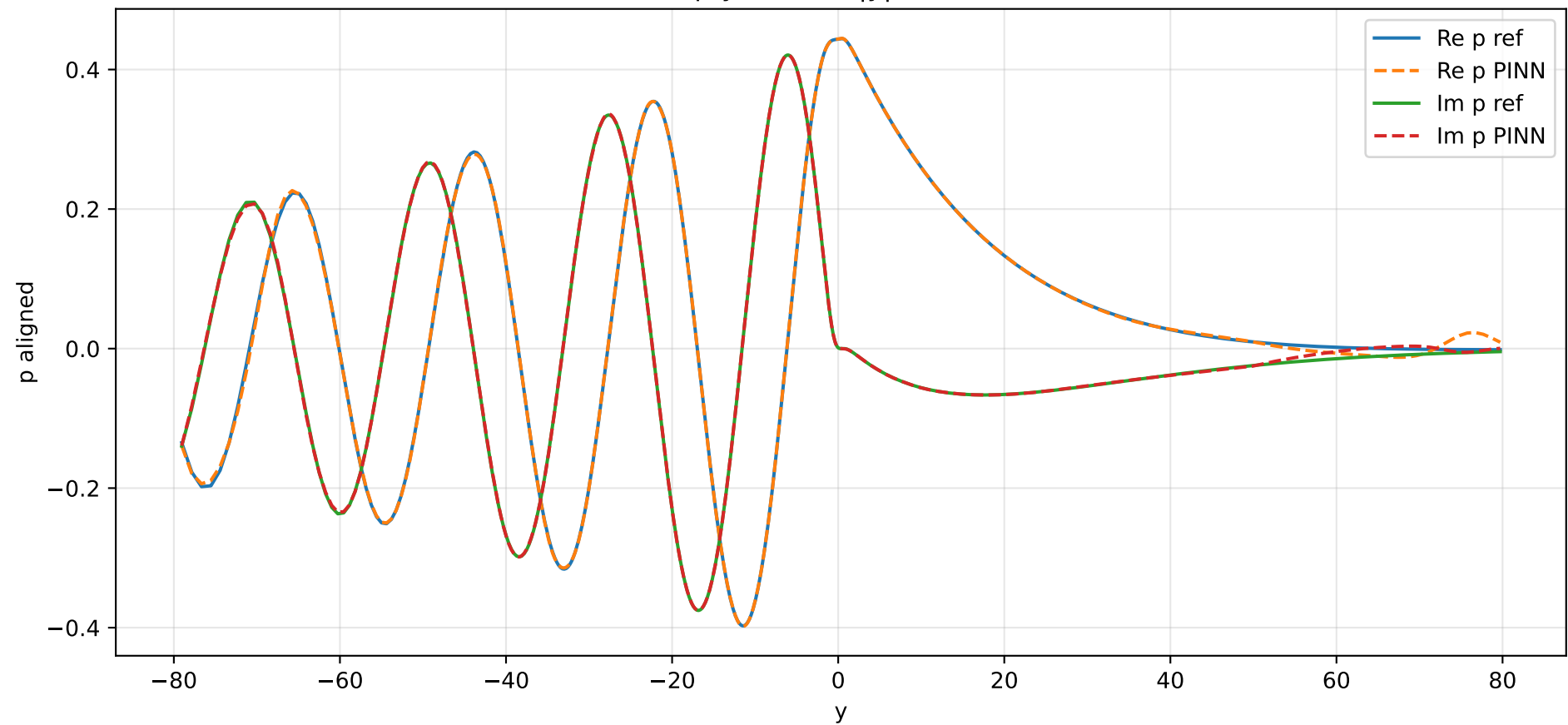
right_y>120: p_rel=0.5903, q_rel=2.557, max|p_pred|=0.0003463

left_y<-80: p_rel=0.5918, q_rel=0.5903, max|p_pred|=0.1941

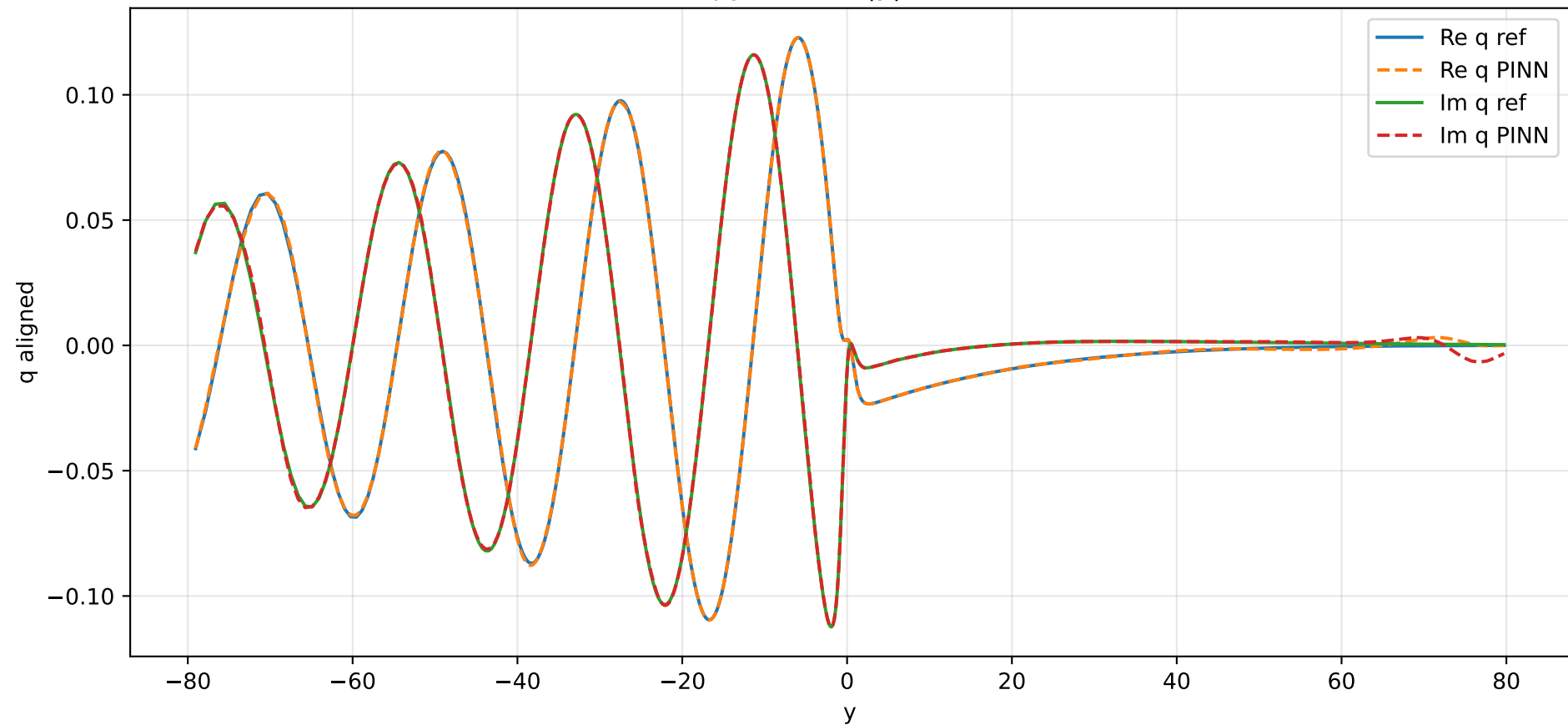
outside_|y|>80: p_rel=0.3613, q_rel=0.3295, max|p_pred|=0.1941

full_domain: p_rel=0.2318, q_rel=0.2223, max|p_pred|=0.4448

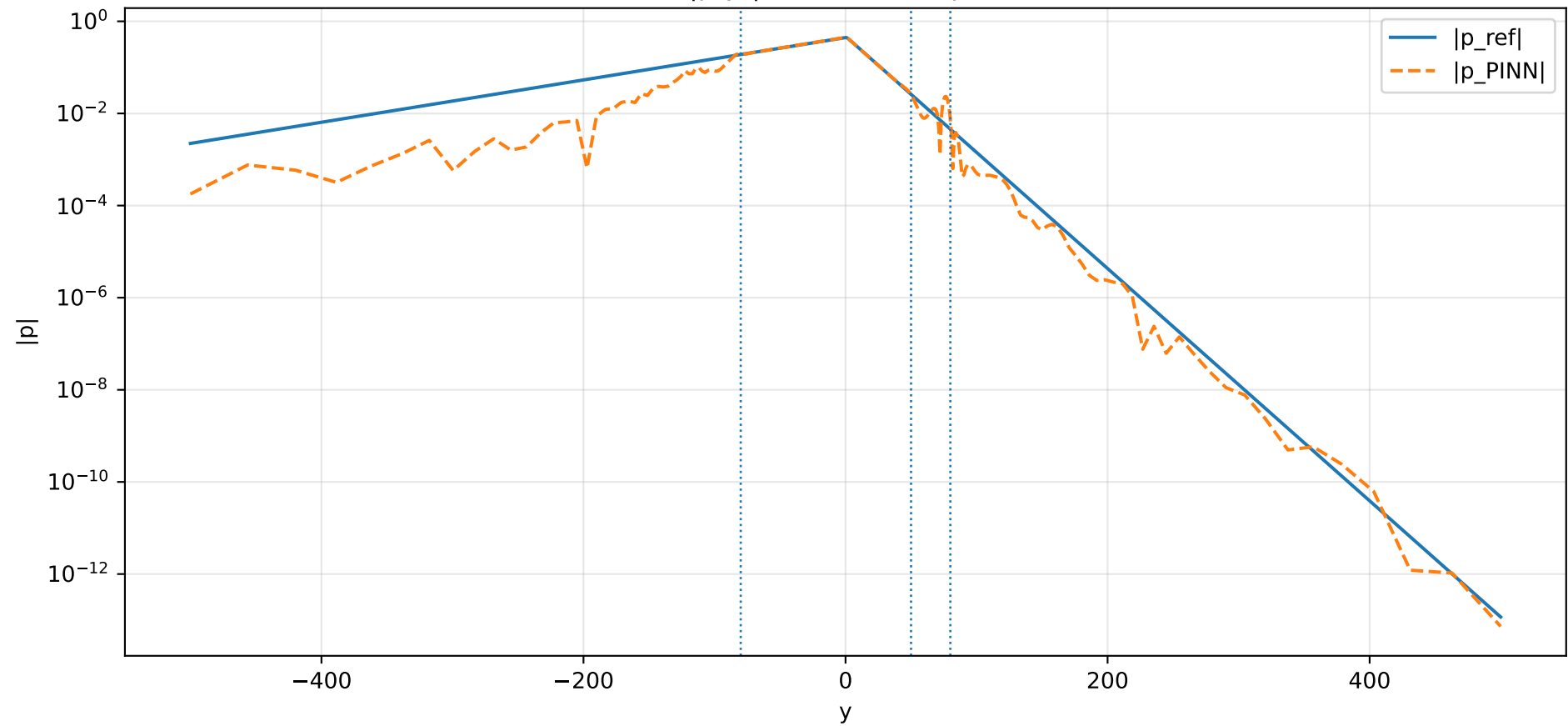
$p(y)$, centre $|y| \leq 80$



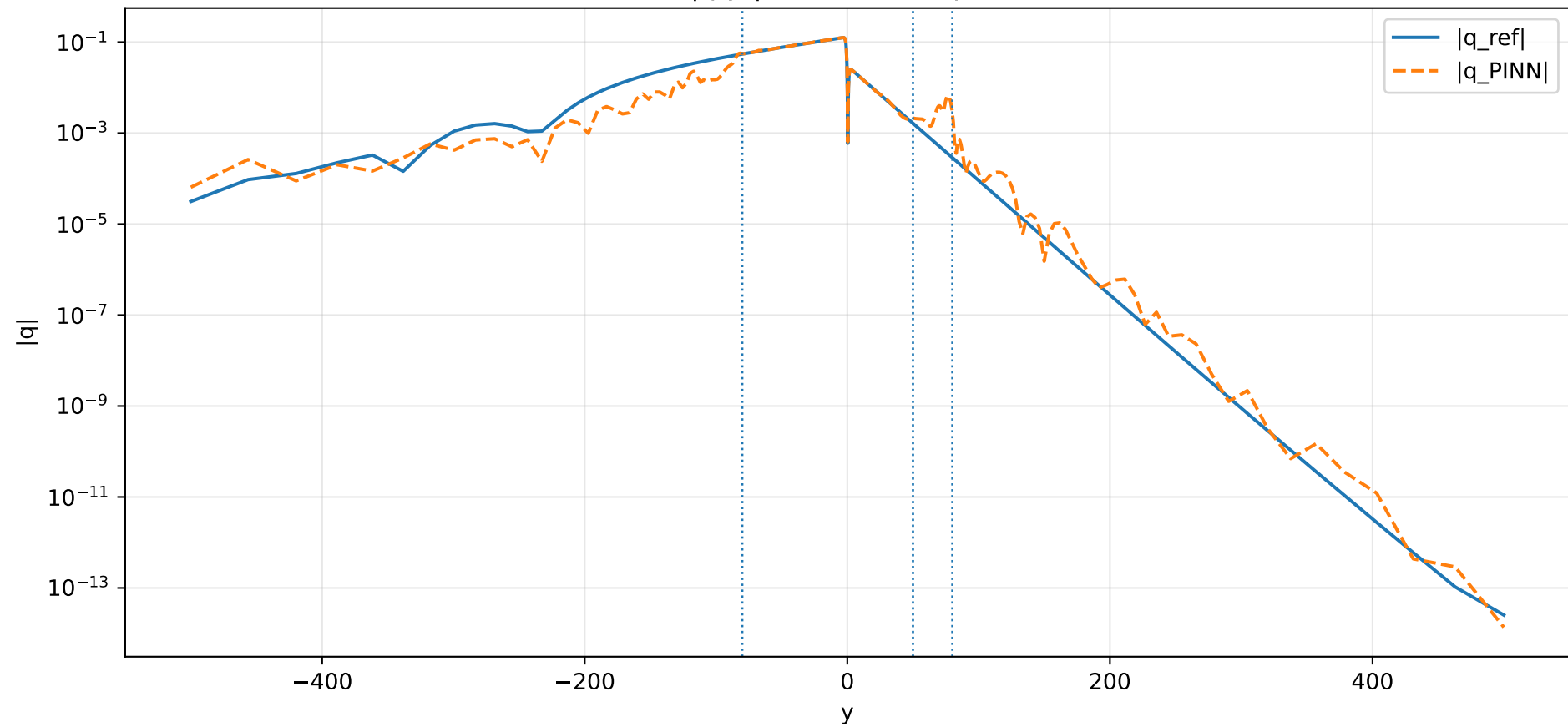
$q(y)$, centre $|y| \leq 80$

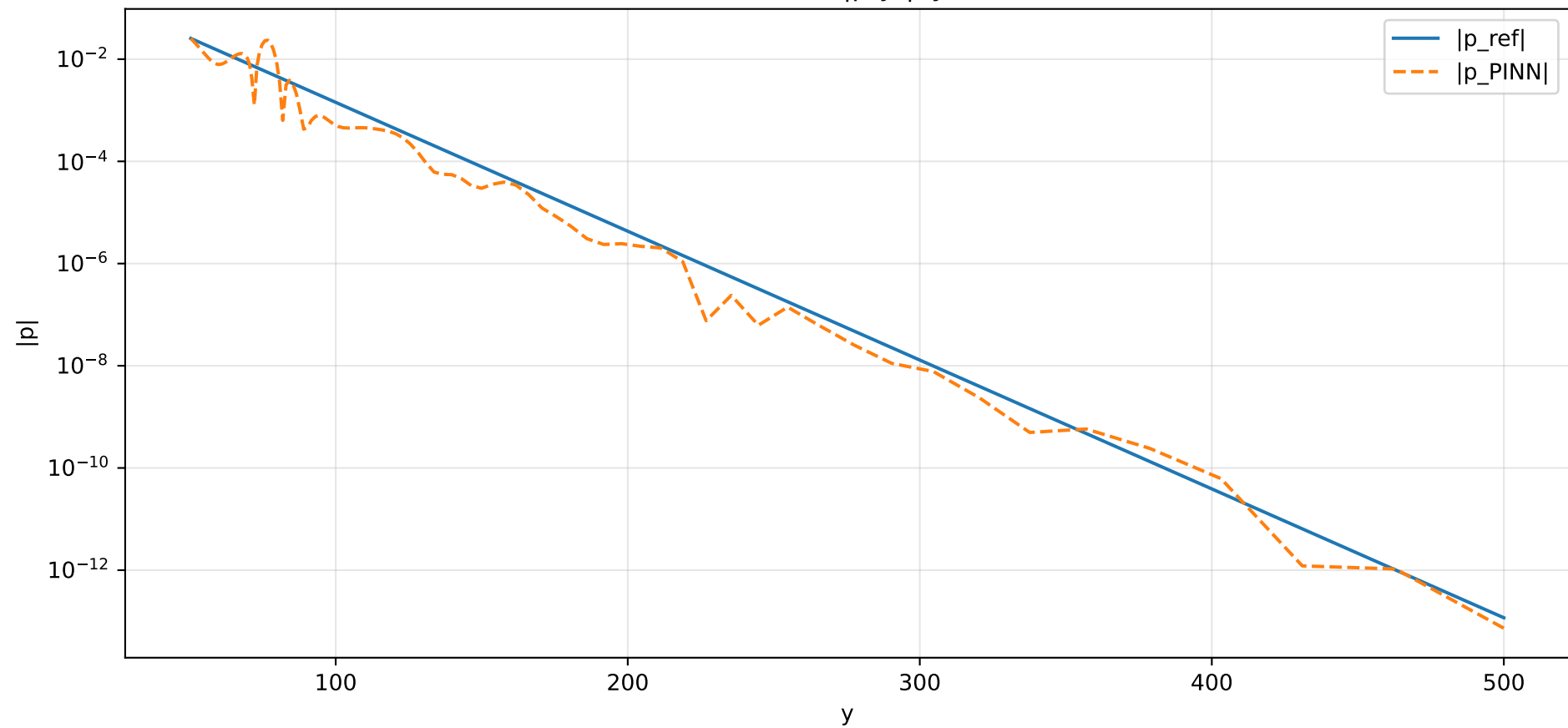


$|p(y)|$ domaine complet

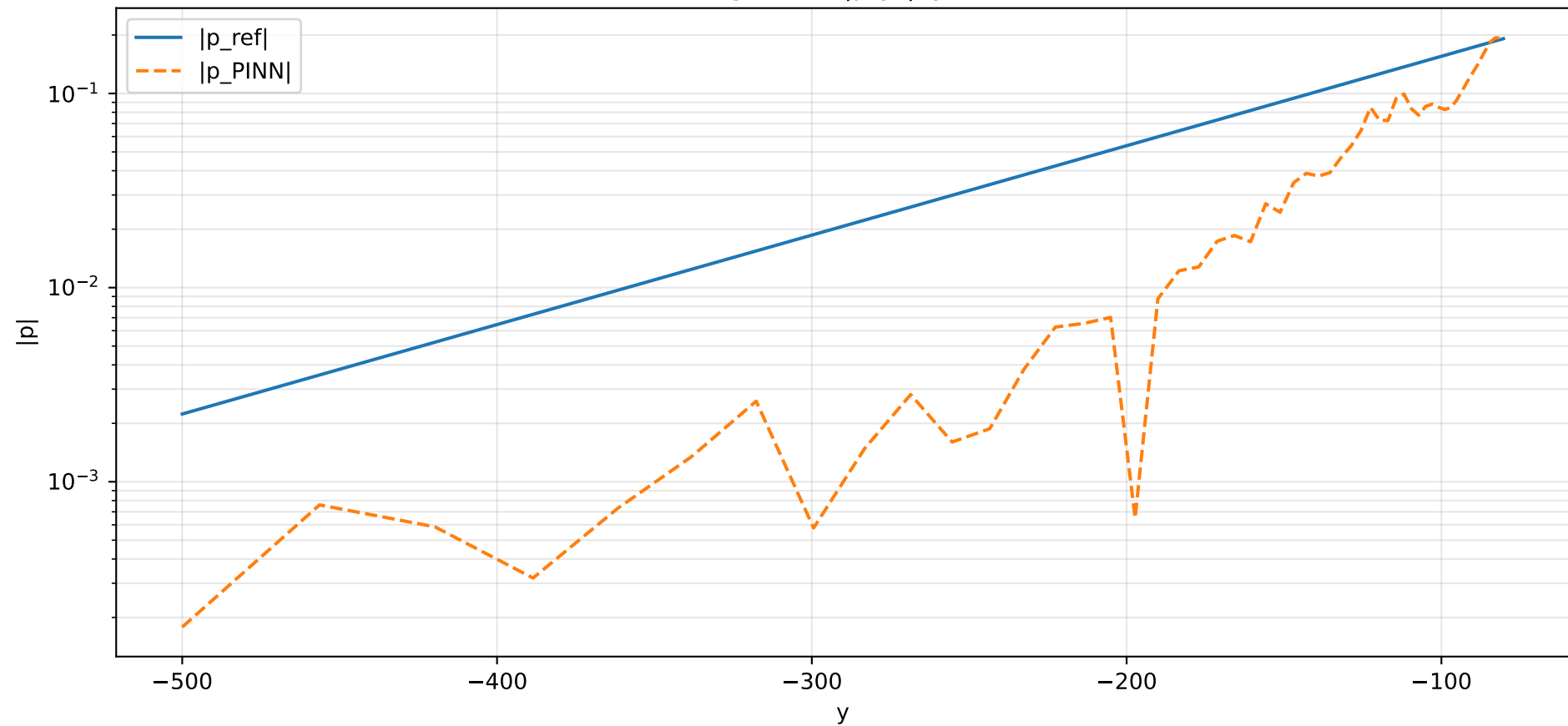


$|q(y)|$ domaine complet

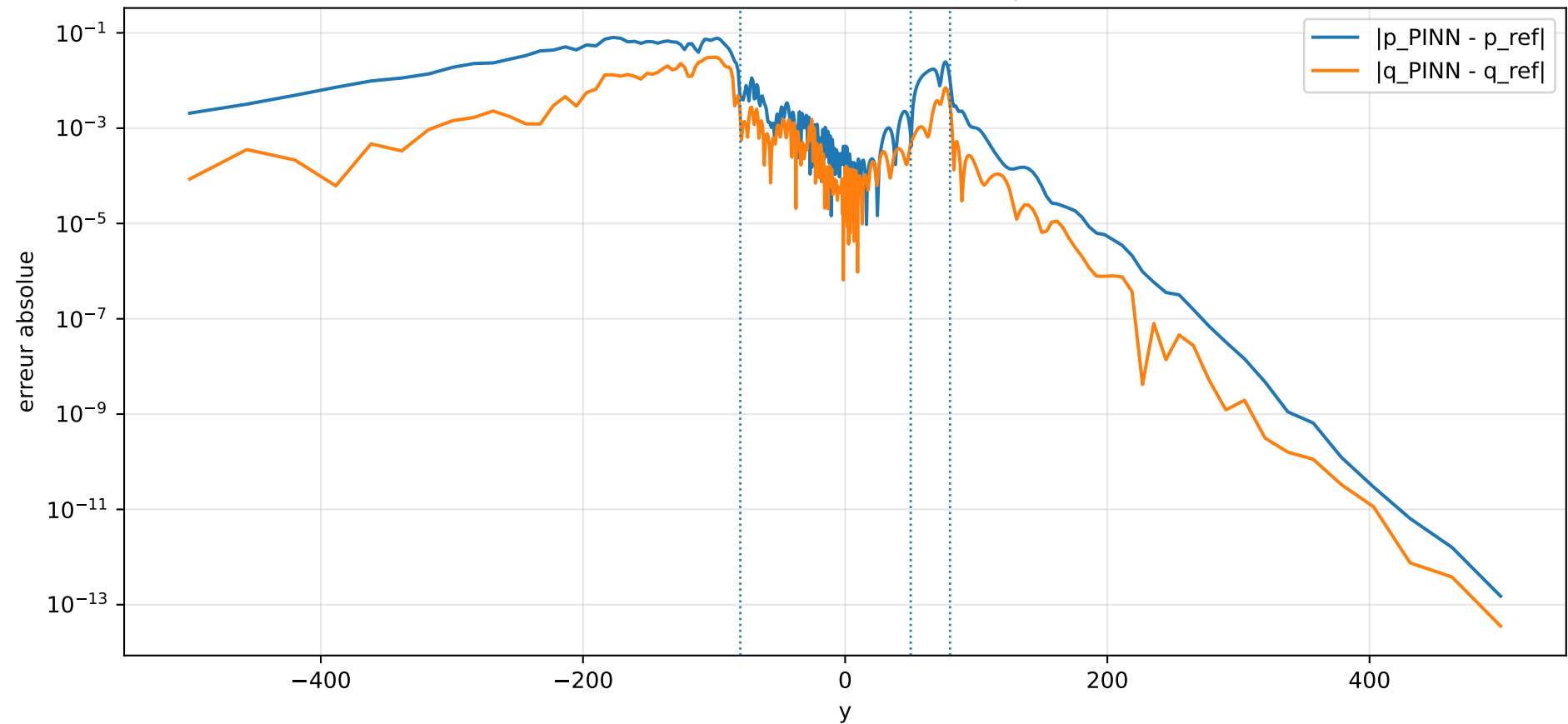


Queue droite, $|p(y)|$, $y \geq 50$ 

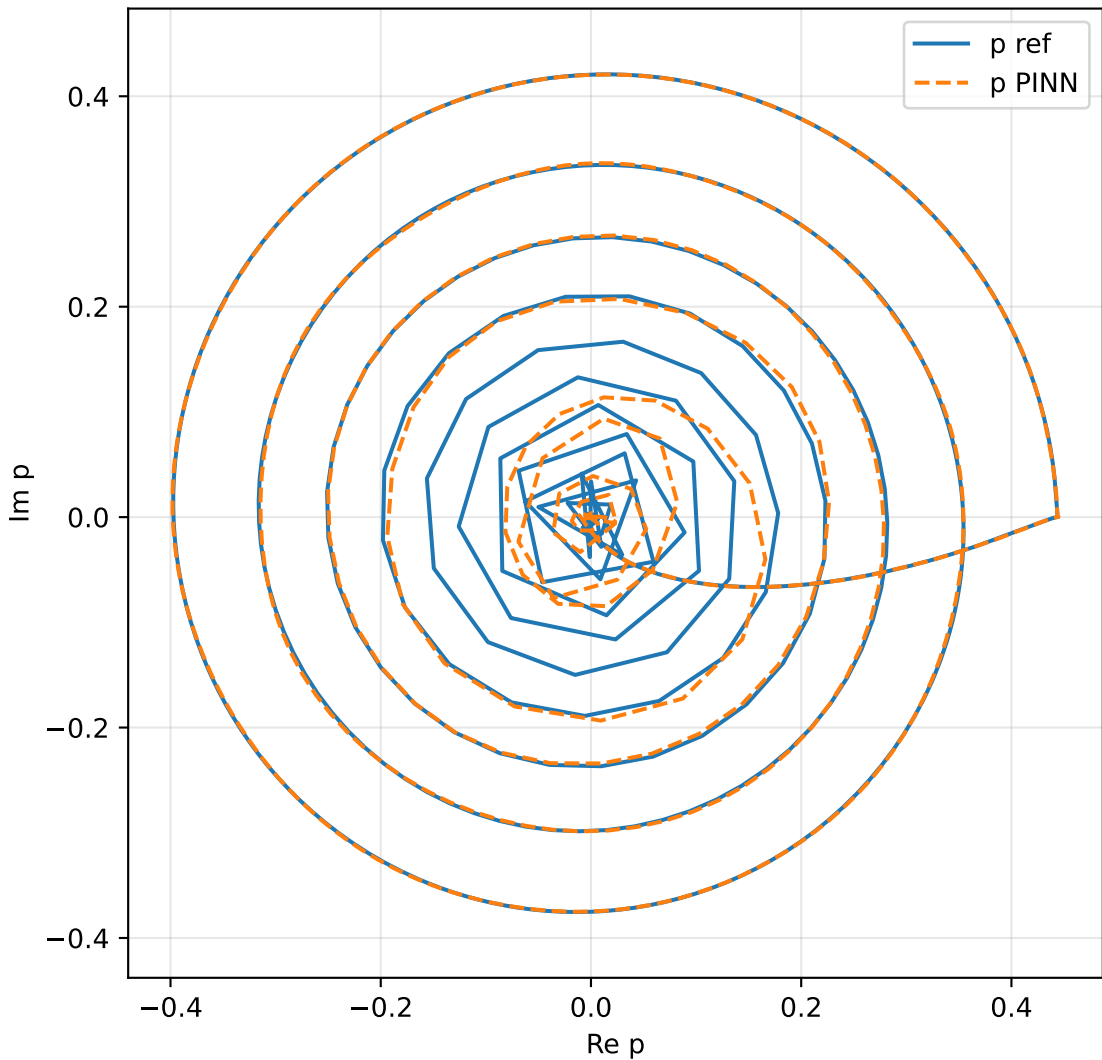
Queue gauche, $|p(y)|$, $y \leq -80$



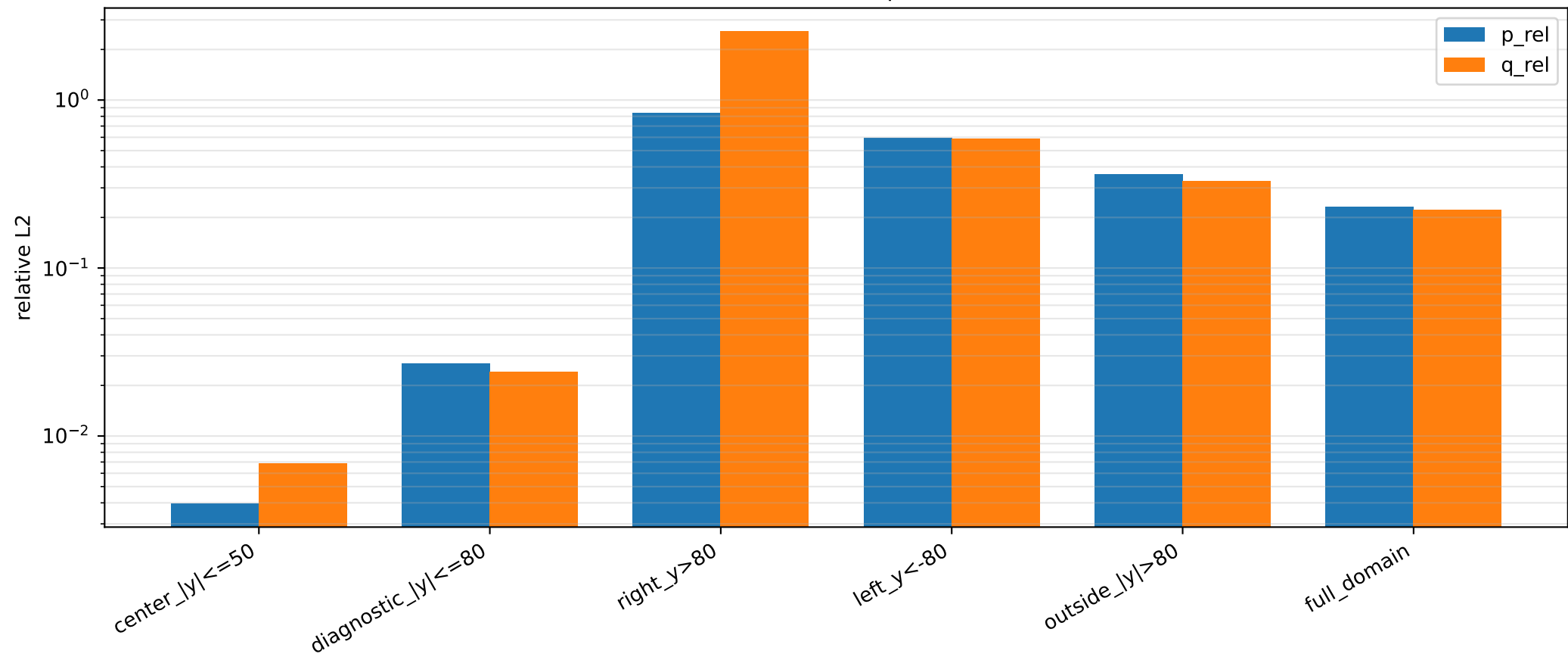
Erreur absolue domaine complet



Trajectoire complexe $p(y)$



Erreurs relatives par zone



Compatibilité $q \approx p_y$, centre $|y| \leq 80$

